

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building an AI clinical decision support app (Next.js + Firebase/GCP) that turns unstructured clinical evidence into structured, guideline-backed medication recommendations inside the EHR prescribing workflow.
- Extending a **Firebase Cloud Functions** backend that ingests multiple clinical data sources, **RxNorm** normalization, and PubMed/guideline evidence into structured results, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility, keeping model output validated against source evidence rather than free-form.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building an embedding-based text-analysis platform (Python, **FastAPI**, sentence-transformers, React) that ingests messy research corpora and turns them into structured, analysis-ready measurements.
- Built versioned data infrastructure, a YAML **model registry**, and a structured **data-quality** warning system with deterministic language detection that flags data issues and reports uncertainty instead of guessing; grew the backend suite to **40 hermetic tests** in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus **REST** integrations and aggregation queries; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability across the multi-tenant eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage LLM pipeline that turns open-ended, unstructured queries into **structured entity tables** via typed retrieval planning, deterministic extraction with LLM fallback, and output verification, with **cell-level provenance** and provider routing/fallback across cloud LLM backends; hardened with **150+ tests**.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an evaluation agent that proposes queries and labels, benchmarks them against multiple retrieval systems, and accepts only updates that improve discriminative power, with anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection).

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash

LLM & Agents: LLM pipelines, agentic systems, provider routing / fallback, prompt iteration, RAG, evaluation

ML & Data: PyTorch, sentence-transformers, scikit-learn, NumPy, pandas, Matplotlib, Hugging Face Transformers

Retrieval / Search: RRF, cross-encoder reranking, MMR, embedding search, cell-level provenance

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, Linux, CI/CD

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.